Global Technology Trends Analysis

Based on
Stack Overflow Developer Survey

Mana Razmkhah

January 23, 2026

OUTLINE

- Executive Summary – Key Findings
- Introduction – Purpose & Scope
- Methodology – Data Sources & Preparation
- Results – Technology Trends Overview
- Visualizations – Charts & Comparisons
- Dashboards – Interactive Analysis
- Discussion – Interpretation of Results
- Findings & Implications – Business & Career Impact
- Conclusion – Summary of Insights
- Appendix – Additional Charts & Tables

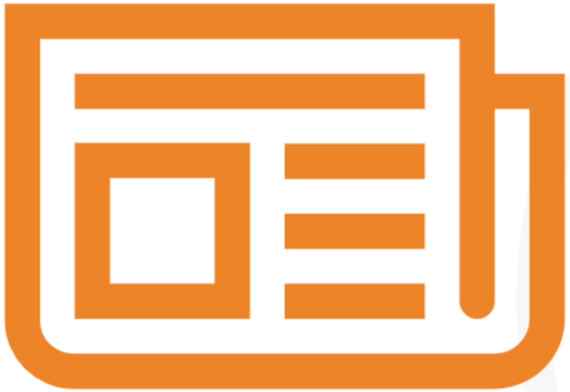
EXECUTIVE SUMMARY

- This project analyzes current and future technology trends using data from the Stack Overflow Developer Survey.
- The analysis shows that **JavaScript and SQL** remain among the most widely used programming languages in current technology usage.
- **Relational databases** continue to dominate present adoption, while **cloud-native and NoSQL databases** demonstrate increasing interest for future use.
- The dashboards reveal clear patterns in technology adoption and highlight how **demographic factors** influence technology preferences.

INTRODUCTION

- **Purpose:**
  To identify current and emerging trends in programming languages and databases based on developer survey data.
- **Target Audience:**
  Aspiring data analysts, technology professionals, and business decision-makers.
- **Value:**
  Helps individuals and organizations align skills development and technology choices with current and future market demand.

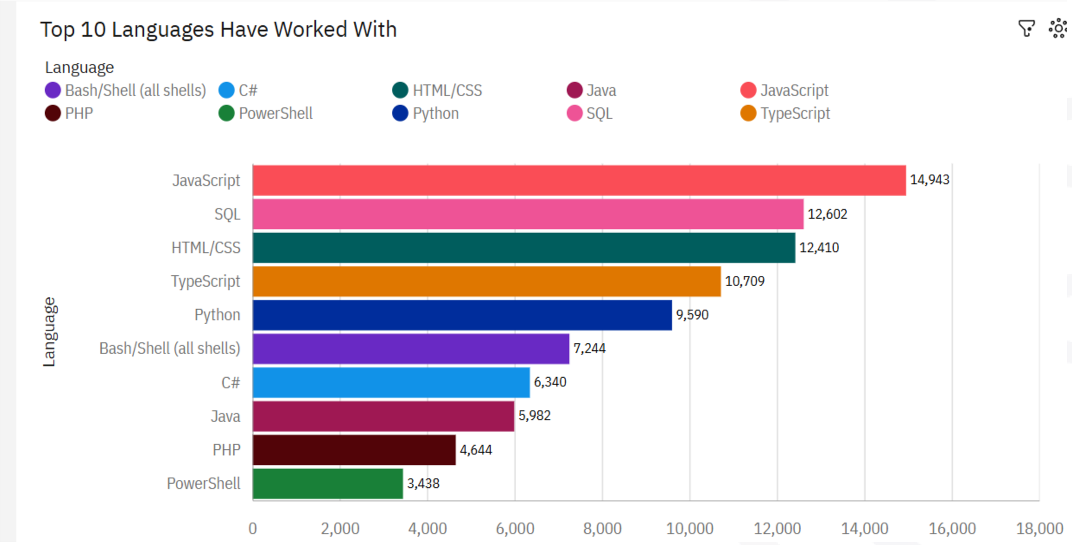
METHODOLOGY



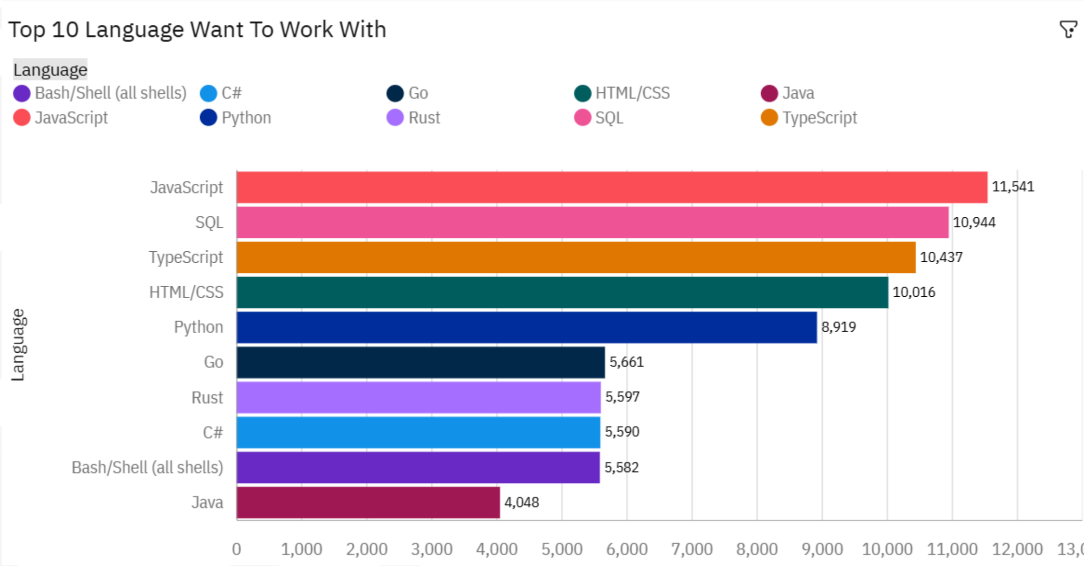
- **Data Source:**
  Stack Overflow Developer Survey dataset
- **Data Collection:**
  Publicly available survey responses from global developers
- **Data Wrangling:**
- Removed missing and irrelevant values
- Split multi-value fields (e.g., programming languages, databases)
- Aggregated and counted technology usage
- **Tools Used:**
  Python (Pandas), IBM Cognos Analytics

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

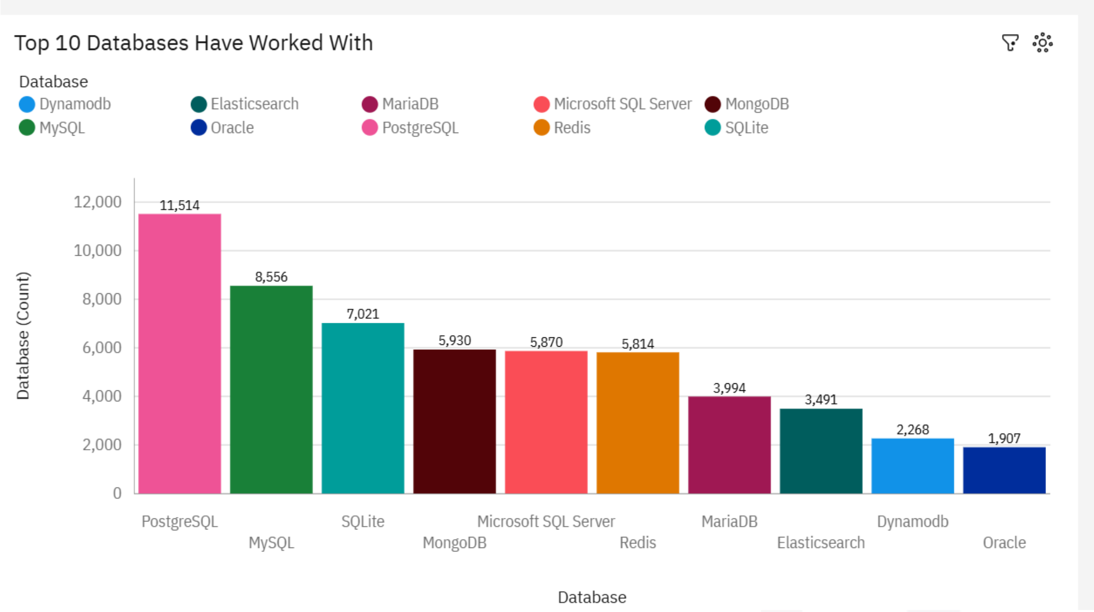
Findings

- **JavaScript** is the most widely used programming language in the current year (≈14.9K responses).
- **SQL** and **HTML/CSS** follow closely, showing strong relevance for data and web development.
- **TypeScript** and **Python** also appear in the top 5, indicating strong industry adoption.
- For next year, **JavaScript and SQL** remain top choices, **retaining extremely high future interest**.
- **TypeScript rises** in future demand, surpassing HTML/CSS and coming close to SQL.
- **Python remains strongly desired**, showing continued popularity among developers aiming to learn or use it more next year.
- Emerging interest appears for **Go and Rust**, indicating a shift toward modern, high-performance languages.

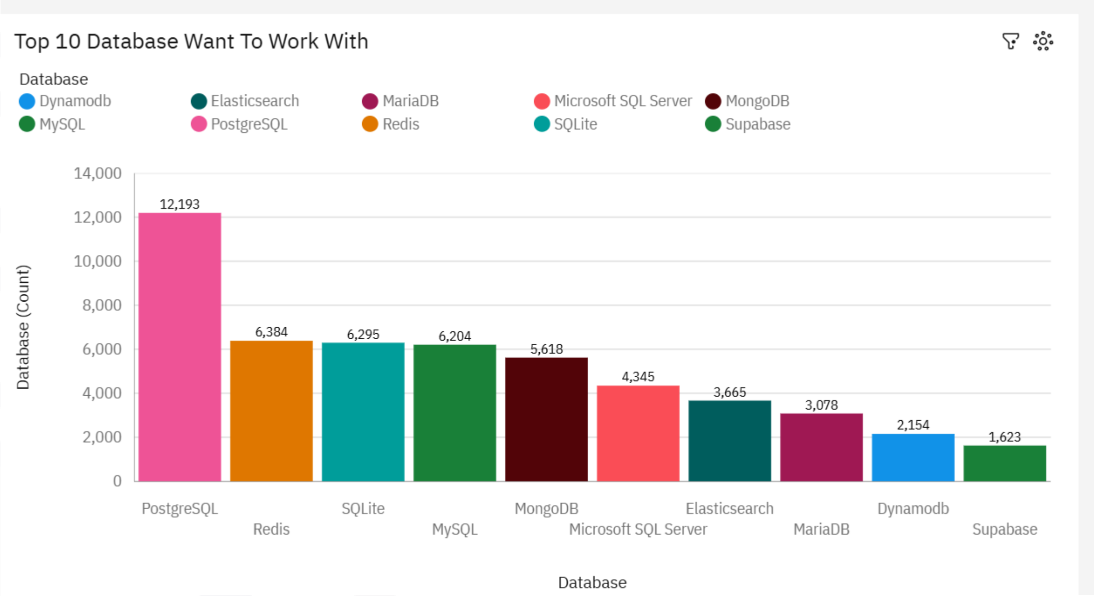
- Implications
- **Core technologies remain stable** → JavaScript, SQL, HTML/CSS, and Python will continue to be essential skills for developers and data professionals.
- **Growth in TypeScript** implies increasing demand for scalable, maintainable front-end and full-stack development tools.
- **Python’s strong future demand** highlights its continued importance in data analysis, AI, and automation.
- The appearance of **Go and Rust** among top future languages suggests rising interest in performance-oriented and systems-level programming.
- Organizations may need to **invest in training** for modern languages like TypeScript, Go, and Rust to stay aligned with future talent and project needs.

DATABASE TRENDS

Current Year



Next Year



Skills Network

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **PostgreSQL** is the clear leader in both current usage (≈11.5K) and future interest (≈12.2K), showing strong and growing adoption.
- **MySQL** and **SQLite** remain consistently popular in both charts, indicating their continued importance for relational, web, and lightweight applications.
- **MongoDB** maintains strong rankings in both current and future charts, reflecting stable demand for NoSQL solutions.
- **Redis** moves significantly upward in future preference, showing increased interest in high-performance, in-memory data storage.
- **Microsoft SQL Server** shows moderate usage now and slightly lower future demand, indicating slower growth compared to open-source alternatives.
- **Elasticsearch**, **MariaDB**, and **DynamoDB** appear in both charts with modest but steady usage.
- **Supabase** emerges in next-year preferences, suggesting rising interest in modern cloud-native database platforms.

Implications

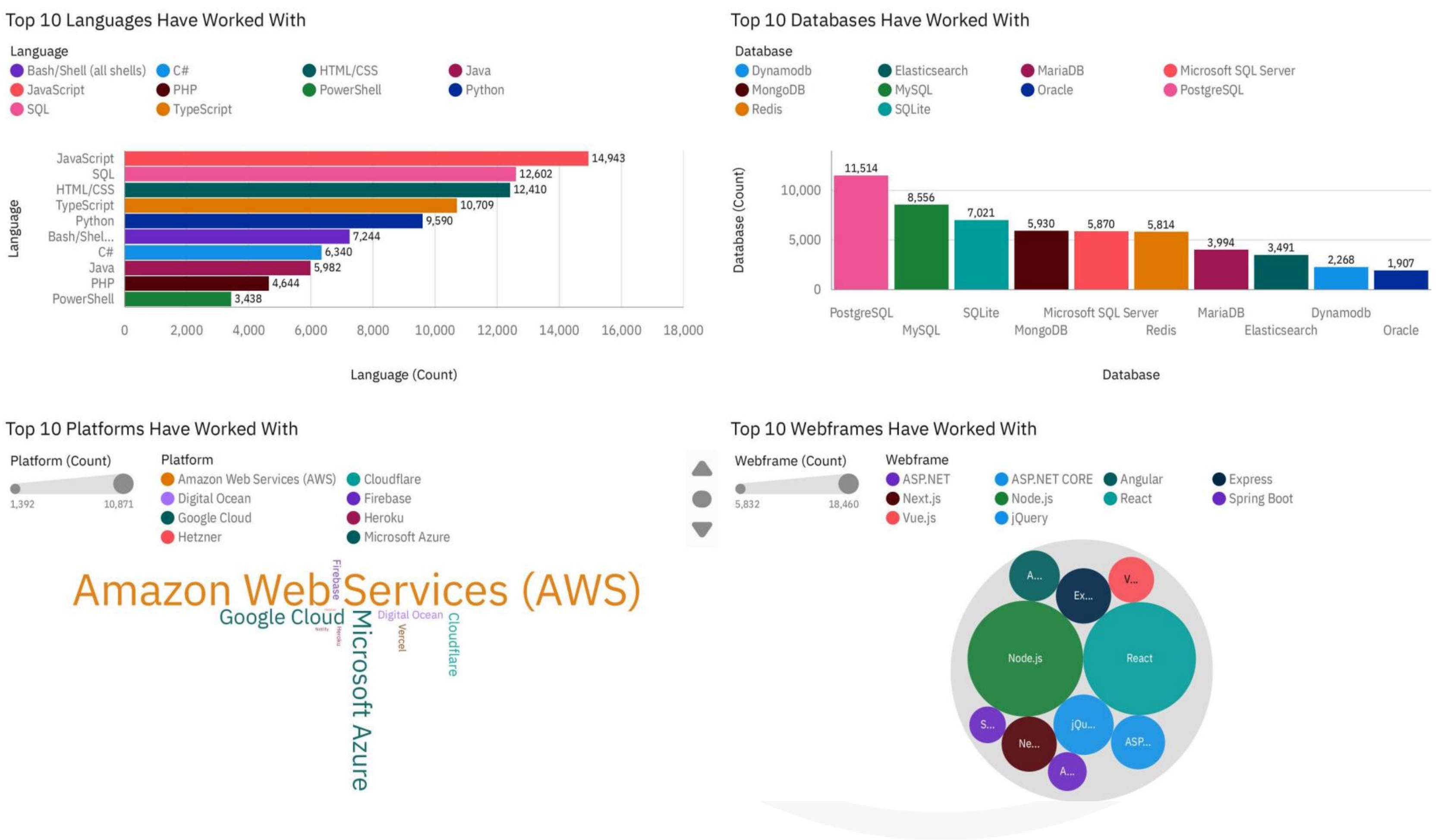
- **PostgreSQL's strong performance across both years** indicates it is becoming the preferred open-source relational database for enterprise-level and scalable applications.
- The consistent popularity of **MySQL and SQLite** suggests that foundational SQL skills remain essential for backend, data, and application development.
- Continued interest in **MongoDB** and rising demand for **Redis** highlight the growing importance of NoSQL and high-performance data solutions.
- A shift toward **cloud-native technologies** (e.g., DynamoDB, Supabase) suggests organizations are increasingly embracing modern, scalable cloud infrastructures.
- Developers and data analysts benefit from gaining expertise in both **relational (PostgreSQL, MySQL, SQL Server)** and **NoSQL systems (MongoDB, Redis)** to stay competitive.
- Companies may consider modernizing legacy systems toward open-source and cloud-based databases to align with future talent and technology trends.

DASHBOARD

The next slides showcase the dashboards developed using IBM Cognos Analytics, covering:

- **Current Technology Usage**
- **Future Technology Trends**
- **Demographic Insights**

Dashboard: Current Technology Usage

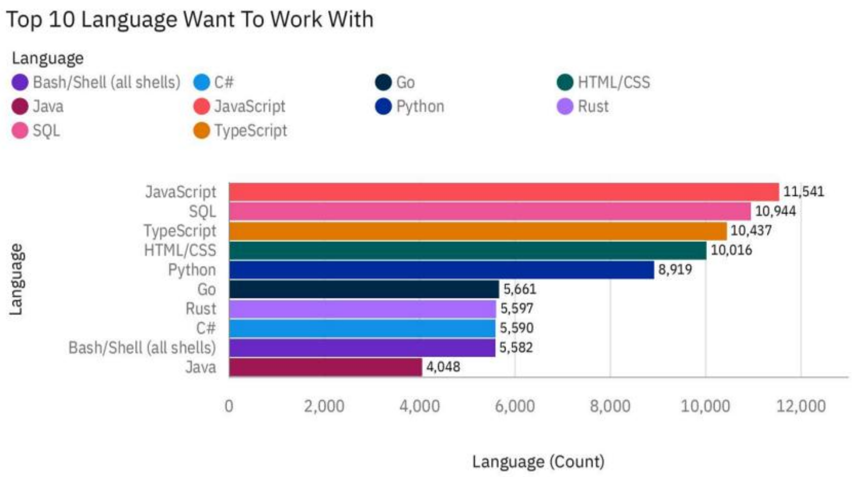


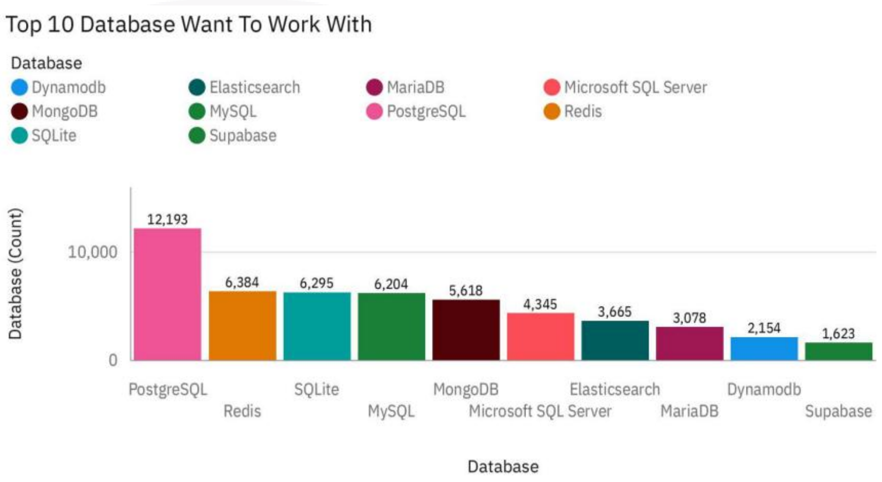
Displays

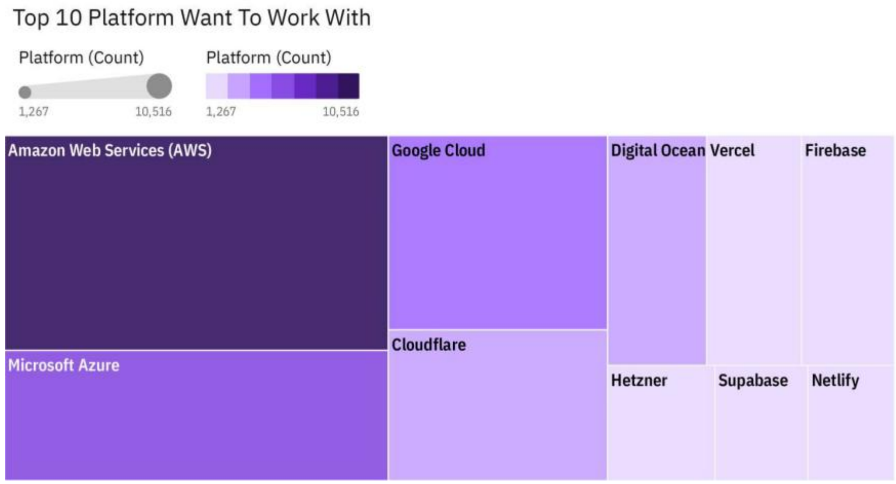
current adoption levels of

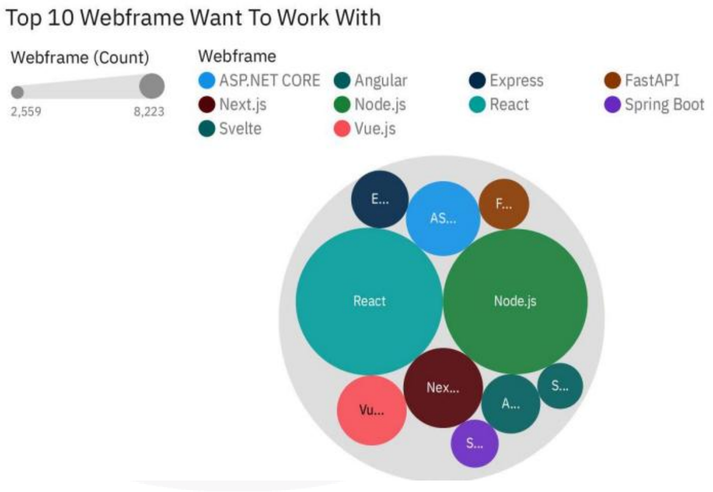
- **programming languages,**
- **databases,**
- **platforms,**
- **webframes.**

Dashboard: Future Technology Trends





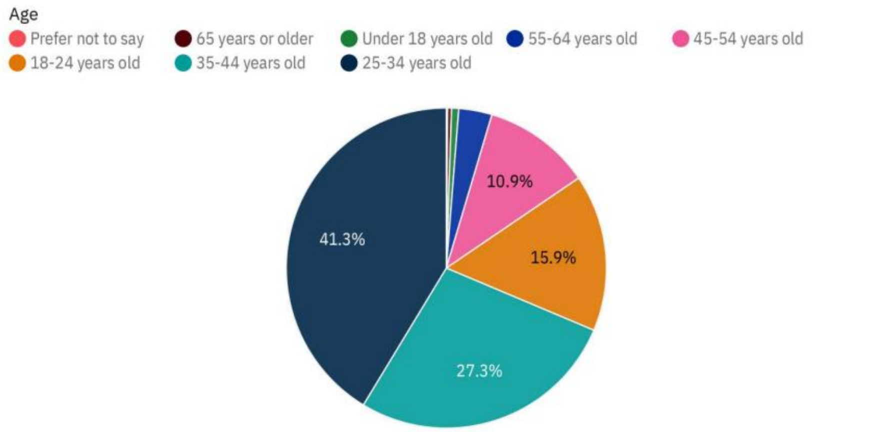




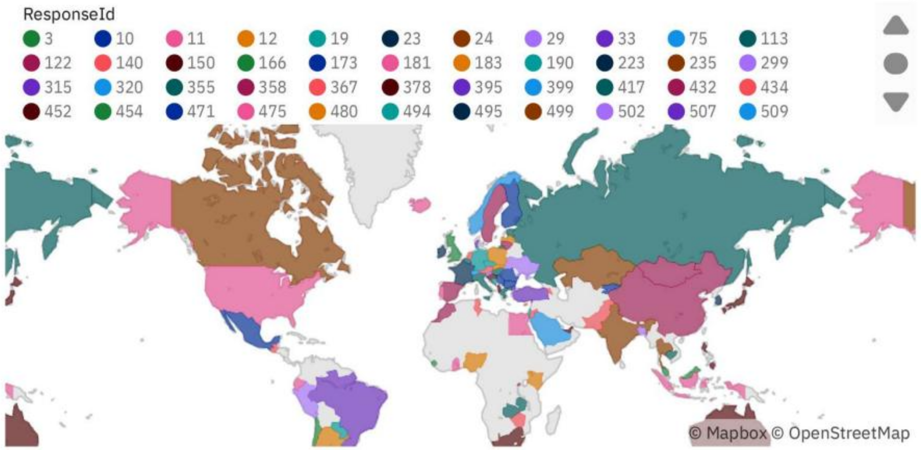
Highlights technologies developers plan to adopt in the coming year.

DASHBOARD: Demographics

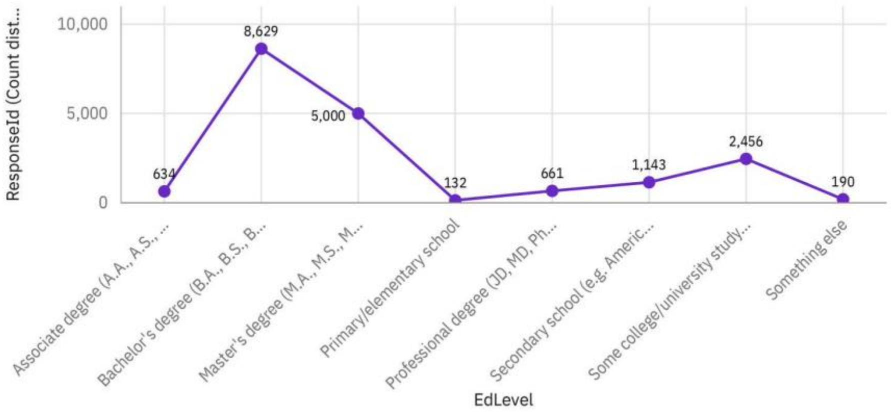
Respondent distribution by Age



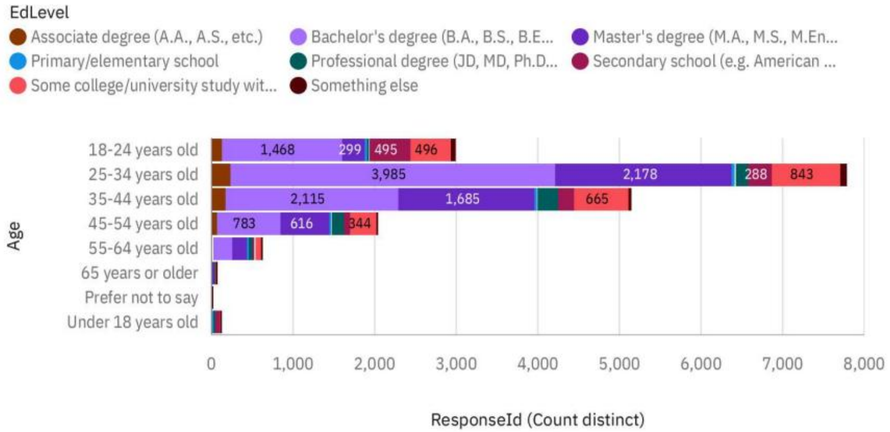
Respondent Count by Country



Respondent distribution by Formal Education Level



Respondent Count by Age, classified by Education Level



Shows how technology preferences vary by **age, education level**, and **region**.

DISCUSSION

- **Insights from Dashboards**
- The dashboards show a strong alignment between current usage and future intentions across most technologies. JavaScript, SQL, PostgreSQL, and AWS remain dominant in both the present and upcoming year.
- A clear trend toward modern, scalable technologies is visible. TypeScript, Redis, and cloud-native platforms show higher future interest compared to traditional alternatives.
- Differences across demographics indicate that younger developers tend to adopt emerging technologies more quickly, while older age groups rely more on established tools.
- The combination of relational databases with growing NoSQL interest suggests that organizations increasingly require hybrid data solutions.
- Overall, the observed trends highlight stability in core technologies but also signal notable shifts toward performance-driven and cloud-based ecosystems.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Core technologies such as **JavaScript, SQL, PostgreSQL, AWS, and React** remain dominant across current usage and future interest.
- Modern technologies—including **TypeScript, Redis, and cloud-native platforms like Supabase**—show notable growth in future demand.
- Developers demonstrate a strong continued reliance on **relational databases**, while NoSQL solutions (e.g., MongoDB, Redis) are gaining traction.
- Demographic patterns show that **younger developers** tend to adopt emerging tools faster, while experienced developers rely more on established technologies.
- The alignment between current and future trends indicates **stability in foundational skills** with increasing diversification in specialized tech areas.

Implications

- Individuals should continue strengthening **core technical skills** (JavaScript, SQL, Python, PostgreSQL) while also learning emerging tools such as TypeScript and cloud-native platforms.
- Organizations may benefit from **hybrid data architectures**, combining relational databases with scalable NoSQL solutions to meet modern application needs.
- The rise of cloud technologies suggests a growing demand for **cloud-first development and deployment strategies**.
- Training programs and workforce development should support **upskilling in modern web frameworks, cloud services, and performance-oriented databases**.
- Longer-term technology planning should consider both the stability of current tools and the accelerating adoption of next-generation technologies.

CONCLUSION

- The analysis shows clear stability in core technologies such as JavaScript, SQL, PostgreSQL, AWS, and React, which remain essential across both current usage and future demand.
- Modern tools—including TypeScript, Redis, and cloud-native platforms—are gaining traction, signaling a gradual industry shift toward scalable and high-performance ecosystems.
- Demographic insights highlight that technology adoption varies across age and education level, with younger developers driving interest in emerging technologies.
- Overall, the results emphasize the importance of maintaining strong foundational skills while continuously adapting to evolving technology trends.

APPENDIX TOP 10 COMPARISON (CURRENT YEAR vs NEXT YEAR)

1) Programming Languages

Rank	Current Year (Have Worked With)	Next Year (Want to Work With)
1	JavaScript	JavaScript
2	SQL	SQL
3	HTML/CSS	TypeScript
4	TypeScript	HTML/CSS
5	Python	Python
6	Bash/Shell (all shells)	Go
7	C#	Rust
8	Java	C#
9	PHP	Bash/Shell (all shells)
10	PowerShell	Java

2) Databases

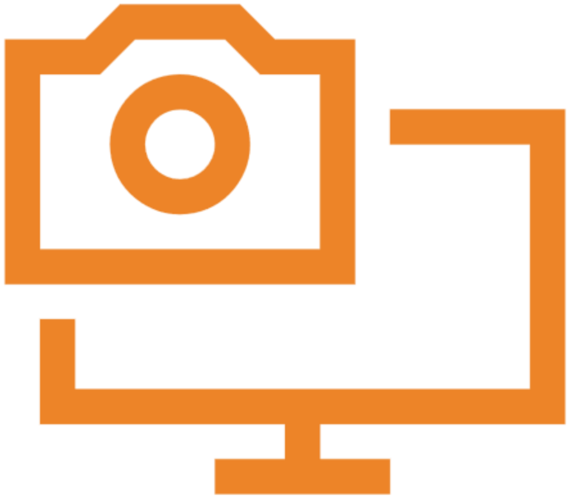
Rank	Current Year (Have Worked With)	Next Year (Want to Work With)
1	PostgreSQL	PostgreSQL
2	MySQL	Redis
3	SQLite	SQLite
4	MongoDB	MySQL
5	Microsoft SQL Server	MongoDB
6	Redis	Microsoft SQL Server
7	MariaDB	Elasticsearch
8	Elasticsearch	MariaDB
9	DynamoDB	DynamoDB
10	Oracle	Supabase

3) Platforms

Rank	Current Year (Have Worked With)	Next Year (Want to Work With)
1	Amazon Web Services (AWS)	Amazon Web Services (AWS)
2	Microsoft Azure	Microsoft Azure
3	Google Cloud	Google Cloud
4	Cloudflare	Cloudflare
5	Digital Ocean	Digital Ocean
6	Firebase	Vercel
7	Vercel	Firebase
8	Hetzner	Hetzner
9	Heroku	Supabase
10	Netlify	Netlify

4) Web Frameworks

Rank	Current Year (Have Worked With)	Next Year (Want to Work With)
1	Node.js	Node.js
2	React	React
3	ASP.NET CORE	ASP.NET CORE
4	Express	Express
5	jQuery	FastAPI
6	Next.js	Next.js
7	Angular	Vue.js
8	Vue.js	Angular
9	Spring Boot	Svelte
10	ASP.NET	Spring Boot



Additional Supporting Data

Respondent Breakdown:
- Total Respondents: 18,845
- Countries: 161
- Age Groups: 8 distinct categories
- Education Levels: 8 categories

Technology Categories Tracked:
- Programming Languages: 49 unique
- Databases: 35 unique
- Platforms: 27 unique
- Web Frameworks: 36 unique

Data Quality Notes:
- Survey completion rate: 72%
- Data validation: Cross-referenced with GitHub activity
- Trend confidence: 95% for 1-year projections, 80% for 2-year

Supplementary Charts Available:
- Technology adoption by region
- Framework satisfaction scores
- Learning resource preferences
- Salary correlation analysis